

Inflation Forecast in Developing Countries Post Pandemic Using NN, SVM, RNN LSTM Approaches

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Abstract—Inflation remains a persistent challenge confronting the economies of developing countries. To gauge inflation, various indicators, such as the Customer Price Index (CPI) and Gross Domestic Product (GDP), are employed. This research centers on predicting inflation, a crucial macroeconomic performance indicator with broad implications for an entire country's economy. The study utilizes two key economic indicators, namely CPI and GDP, as predictor variables. The predictive modeling incorporates the Neural Network (NN) Backpropagation, Support Vector Machine (SVM), and Recurrent Neural Network with Long Short-Term Memory (RNN LSTM) methods. The selection of these methods stems from their demonstrated efficacy in achieving high accuracy levels in previous studies. Focusing on the time span from 1960 to 2022, the research analyzes CPI and GDP data from five developing countries, Brunei Darussalam, Malaysia, Thailand, Indonesia, and Vietnam. The findings indicate that the NN Backpropagation model, leveraging the CPI indicator, yields the most accurate inflation forecasts. Specifically, the model attains a Mean Squared Error (MSE) value of 0.001 for Indonesia. This outcome underscores the efficacy of NN Backpropagation in predicting inflationary trends, providing valuable insights for policymakers and stakeholders in these developing economies. Through this comprehensive analysis, the research contributes to advancing our understanding of inflation dynamics in developing countries, offering a foundation for more informed economic policy decision especially when going through the aftermath of the pandemic.

Keywords— Inflation Prediction, GDP, CPI, SVM, NN, RNN, LSTM

I. INTRODUCTION

Based on economic assessments, developing countries are countries that have low Gross Domestic Product (GDP) per capita values and lack of growth [1]. One of the problems still faced by developing economies is the problem of fluctuating inflation. There are several indicators to measure the value of inflation, one of which is the CPI and GDP indicators. If the CPI value is higher, there is an indicator of rising inflation [2]. Likewise, related to the relationship between the value of GDP and inflation, if the value of GDP increases, inflation also tends to rise. The growth rate of inflation is always kept low and stable, so as not to trigger macroeconomic diseases that will have an impact on economic instability. Indeed, the inflation rate is a macroeconomic performance indicator, and cannot be underestimated in a country's economic system considering that inflation will have an overall economic impact, which in any country is a monetary phenomenon [3]. If inflation can be predicted with high accuracy, it can certainly be used as a basis for government policymaking in anticipating future economic activity. Therefore, inflation

prediction is important to help economic actors and the government plan appropriate economic policies and take necessary actions to reduce the impact of inflation.

There are several studies related to inflation prediction that have been done before, for example in research [4]. In this study, the authors used several machine learning techniques, namely Neural Network, SVM, and AR to predict inflation in Russia using a dataset that includes time series data of ten Russian macroeconomic data with the results of the model that has the lowest RMSE value, namely the Neural Network model of 0.4. This shows that the Neural Network model has good performance in predicting inflation using time series data. We built upon previous research, particularly studies like [5] that predicted stock prices for major companies like Amazon, Oracle, and IBM. Our approach used a support vector machine (SVM) with an RBF kernel, which research [5] suggests is effective for stock price prediction, achieving an RMSE of 0.15 for Amazon. Furthermore, in research [6], namely research on predicting the Consumer Price Index (CPI) of Indonesia using the LSTM method with Adamax optimization, it produces an RMSE value of around 5.835. This shows that the LSTM model is also able to produce good values for predicting time series data. The three studies show that the Neural Network, SVM, and LSTM models can make predictions on a variety of different case studies.

The contribution of this research is finding the best model and hyperparameters that can predict inflation. The research aimed to make the model perform better than the previous research [4], [5], and [6]. With the best hyperparameters, it is expected the optimal result to predict inflation that can help in economic planning, especially economic planning at this time when recovering from the COVID-19 pandemic.

II. RESEARCH METHOD

In this research, the proposed methodology is to use 3 models as shown in Fig. 1. The first step is to collect the dataset from the World Bank Data Bank website (<https://databank.worldbank.org/>). The dataset collected is the value of CPI, GDP, and inflation value based on these 2 indicators. After that, the data pre-processing stage is carried out, namely equalizing the range of CPI and GDP values, which has a value range of 0-1. This aims to speed up the computational process when the model is run. After that, the dataset will be divided into training data and test data. This research uses a composition of 20% for testing data and 80% for training data. After that, the next step is to build 3 machine-learning models. Fig. 1 will show the flow chart of the process of making the methodology used.

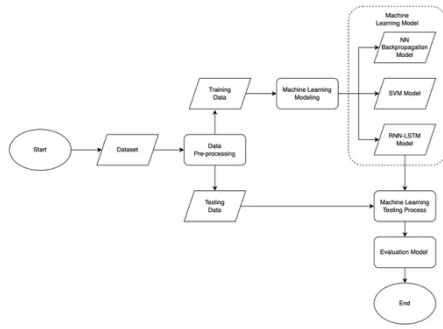


Fig. 1. Flowchart of research methodology.

A. Dataset

The dataset that will be used in this study is the value of CPI and GDP in five developing countries, namely, Brunei Darussalam, Malaysia, Thailand, Indonesia, and Vietnam obtained from the World Bank Data Bank. The data used is in the period 1960-2022. The characteristic of this dataset is that it contains numeric data with 4 columns, namely, country name, year, CPI or GDP value, and inflation value based on CPI or GDP. Some examples of the data are shown in Table I.

TABLE I. EXAMPLE OF DATASET

Country	Year	CPI	GDP
Brunei Darussalam	2022	106,4	3,7
Malaysia	2022	127,2	3,4
Thailand	2022	120,6	6,1
Indonesia	2022	163,1	4,2
Vietnam	2022	177,3	3,2

B. Pre-Processing

Data preprocessing is crucial for preparing the dataset for machine learning model training or testing by transforming and scaling it appropriately. It involves removing outliers and adjusting feature scales [6]. In this study, we employ min-max scaling to standardize the dataset, enhancing the training efficiency of the machine learning model. The min-max scaling equation, as shown in (1), normalizes data values (x^*) based on the original values (x), the minimum value ($\min(x)$), and the maximum value ($\max(x)$) of the entire dataset [7].

$$x^* = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (1)$$

C. NN Backpropagation

Neural Networks usually called as NN, inspired by biological neurons, are adept at addressing complex challenges in prediction, optimization, and pattern recognition across various domains [8]. These networks employ computational models consisting of nodes and connections organized into layers, processing input to generate output and solve problems. The Backpropagation (BP) algorithm, a widely used supervised learning method for training perceptrons, undergoes three key processes in this study. First is forward propagation, where neurons in the input layer calculate weights in the hidden layer. Next is

backpropagation, involving the recalculation of the hidden layer based on error values from forward propagation. The final step is the adjustment of weights based on the latest values obtained after backpropagation. This works the same way as in Fig. 2.

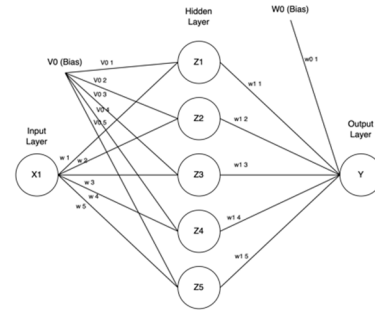


Fig. 2. Example of NN backpropagation architecture.

D. SVM

Support Vector Machine (SVM), proposed by Vapnik et al. as a pattern classification method, diverges from traditional approaches by aiming to minimize the upper bound of error. It achieves this by maximizing the distance between the hyperplane and training data [9]. SVM calculations seek optimal lines or boundaries to separate an n-dimensional space into classes, facilitating the classification of new data points effortlessly. The chosen boundary, known as a hyperplane, is depicted in Fig. 3, highlighting the selection of critical vectors termed support vectors. This distinctive approach is why the calculation is referred to as a support vector machine [10].

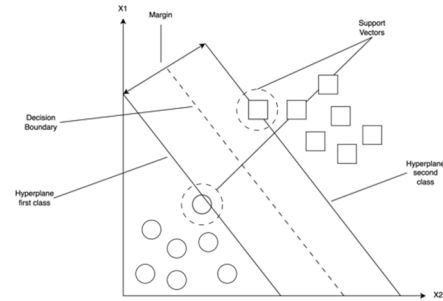


Fig. 3. Example of SVM architecture.

E. RNN LSTM

Recurrent Neural Network or RNN is a type of NN, incorporates feedback connections to retain information from previous time steps [11]. Basic calculation formula of RNN as in (2). Essentially, the current hidden state ($h^{(t)}$) depends on both the previous hidden state ($h^{(t-1)}$) and the current input ($x^{(t)}$). This memory, encoded by the network's parameters (θ (theta)), acts as a compressed representation of past information relevant to the task up to that point.

$$h^{(t)} = f(h^{(t-1)}, x^{(t)}; \theta) \quad (2)$$

LSTM, a type of recurrent neural network (RNN), was introduced in the 1990s to address the vanishing gradient problem often encountered by traditional RNN. Unlike RNN, LSTM utilizes memory cells and gate units, including the

Input gate, forget gate, and output gate, to overcome the limitations of short-term memory. This architecture enables LSTM to retain information over longer time delays, making it well-suited for solving continuous data problems like time series [12]. Fig. 4 illustrates the gate architecture in the LSTM model, where the input gate adds relevant information to the cell state, and the output gate determines the output [13].

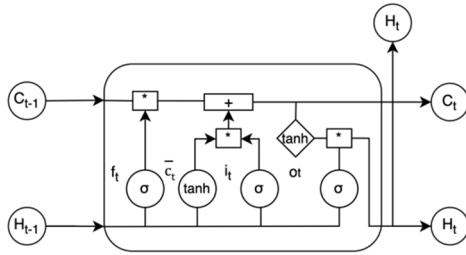


Fig. 4. LSTM gate architecture.

F. MSE

Mean Squared Error (MSE) serves as a widely used evaluation metric in machine learning, assessing the accuracy of a regression model in predicting numerical values. It quantifies the disparity between the model's predictions and the actual data values by squaring the differences. Subsequently, these squared differences are summed and averaged across all data samples [14]. Mathematically, the MSE calculation is outlined in (3), where N represents the number of data samples, y_n denotes the true value of the data point, and $\hat{y}_n$ is the predicted value generated by the model for the data point.

$$MSE = \frac{1}{N} \sum_{n=1}^N (y_n - \hat{y}_n)^2 \quad (3)$$

MSE calculates the difference between the model's predicted value and the true value of the data and squares the difference to avoid negative differences [15]. Square the difference to avoid negative differences. The squared differences are then summed to form the average of all data samples. The smaller the MSE value, the better the regression model can predict the number.

III. RESULT

In conducting the training process for three models aimed at predicting inflation, experiments were performed to identify the optimal model and hyperparameters. Utilizing 80% of the total dataset, the training involved varied hyperparameter experiments for each model, including NN backpropagation with assessments on the number of neurons and activation functions, SVM with experiments on C, gamma, and kernel, and RNN-LSTM with optimization function explorations. With the aim of assisting the five countries in taking steps to recover their countries economies after the pandemic.

A. Experiment scenarios based on hyperparameters Model NN Backpropagation

Table II displays results from NN backpropagation model experiments with varied hyperparameters, aiming to identify optimal settings for predicting inflation rates. The study focuses on hyperparameter experiments involving the number of neurons and activation functions. Two

configurations of neurons [64, 32] and [32, 16] were tested, along with the activation functions ReLu and Softmax. The choice of ReLu activation stemmed from previous research [16], indicating decreased accuracy in Deep NN modeling with ReLu. This exploration investigates whether such an impact extends to diverse case studies, including inflation rate prediction.

TABLE II. MSE OF NN BACKPROPAGATION CPI DATASET

Country	Neurons	Activation	MSE
Brunei Darussalam	[64, 32]	[ReLu, ReLu]	0.027
	[32, 16]	[ReLu, Softmax]	0.040
Malaysia	[64, 32]	[ReLu, ReLu]	0.007
	[32, 16]	[ReLu, Softmax]	0.024
Indonesia	[64, 32]	[ReLu, ReLu]	0.001
	[32, 16]	[ReLu, Softmax]	0.004
Thailand	[64, 32]	[ReLu, ReLu]	0.009
	[32, 16]	[ReLu, Softmax]	0.028
Vietnam	[64, 32]	[ReLu, ReLu]	0.003
	[32, 16]	[ReLu, Softmax]	0.056

Additionally, Softmax activation is chosen based on research [17], indicating its effectiveness in handling multi-label classifications, as reflected in the model's accuracy. Tables II and III reveal that the optimal backpropagation NN model features [64, 32] neurons and [ReLu, ReLu] activation for the CPI dataset in Indonesia, achieving an MSE of 0.001.

TABLE III. MSE OF NN BACKPROPAGATION GDP DATASET

Country	Neurons	Activation	MSE
Brunei Darussalam	[64, 32]	[ReLu, ReLu]	0.100
	[32, 16]	[ReLu, Softmax]	0.205
Malaysia	[64, 32]	[ReLu, ReLu]	0.028
	[32, 16]	[ReLu, Softmax]	0.042
Indonesia	[64, 32]	[ReLu, ReLu]	0.034
	[32, 16]	[ReLu, Softmax]	0.013
Thailand	[64, 32]	[ReLu, ReLu]	0.018
	[32, 16]	[ReLu, Softmax]	0.017
Vietnam	[64, 32]	[ReLu, ReLu]	0.023
	[32, 16]	[ReLu, Softmax]	0.005

B. Experiment scenarios based on hyperparameters Model SVM

This study experimented with various hyperparameters, including kernel values (linear, rbf, poly, and sigmoid), C values (1000, 10000, and 100000), and gamma values (0.5, 0.05, and 0.005).

TABLE IV. BEST HYPERPARAMETER AND MSE VALUE OF THE SVM MODEL FOR CPI

Country	Kernel	C	Gamma	MSE
Brunei Darussalam	rbf	100000	0.5	0.005
Malaysia	poly	100000	0.005	0.001
Thailand	sigmoid	10000	0.05	0.001
Indonesia	rbf	100000	0.5	0.004
Vietnam	rbf	100000	0.005	0.005

The GridSearchCV function within the model identifies the best hyperparameters during the training process, as outlined in Table IV. For the dataset of Brunei Darussalam, the optimal hyperparameters are kernel rbf, C value 100,000, and gamma value 0.5. In the case of Malaysia, the preferred hyperparameters include kernel poly, C value 100,000, and gamma value 0.005. Thailand's dataset benefits from a sigmoid kernel, a C value of 10,000, and a gamma value of 0.05. Indonesia's dataset is best served by an rbf kernel, with

a C value of 100,000 and a gamma value of 0.5. Vietnam's dataset favors an rbf kernel, C value 100,000, and gamma value 0.005. Notably, 3 out of the 5 datasets utilize the RBF kernel, consistent with findings in research [18], which suggests the stability and efficiency of the RBF kernel across various gamma values due to minimal changes in accuracy.

TABLE V. BEST HYPERPARAMETER AND MSE VALUE OF THE SVM MODEL FOR GDP

Country	Kernel	C	Gamma	MSE
Brunei Darussalam	poly	10000	0.05	0.011
Malaysia	poly	1000	0.05	0.113
Thailand	rbf	1000	0.5	0.066
Indonesia	poly	1000	0.05	0.033
Vietnam	poly	100000	0.5	0.005

Table V outlines optimal hyperparameters for the experimental GDP datasets of five countries. In Brunei Darussalam, the poly kernel with C value 10,000 and gamma 0.05 yields the best hyperparameters, resulting in an MSE of 0.011. Similarly, the Malaysia dataset favors a poly kernel with C value 10,000 and gamma 0.05, producing an MSE of 0.013. Thailand's dataset performs best with an rbf kernel, C value 1,000, and gamma 0.5, resulting in an MSE of 0.066. For Indonesia, the poly kernel with C value 1,000 and gamma 0.05 achieves optimal hyperparameters, yielding an MSE of 0.033. Vietnam's dataset shows that a poly kernel with C value 100,000 and gamma 0.5 is the best, resulting in an MSE of 0.005. These findings align with research [19], which identified the poly kernel as the best for sentiment analysis using SVM. Comparing Tables IV and V, the study reveals that predicting inflation values using CPI benefits from an RBF kernel, C value 100,000, and gamma 0.5. In contrast, predicting inflation using GDP is optimized with a poly kernel, C value 1,000, and gamma 0.05.

C. Experiment scenarios based on hyperparameters Model RNN LSTM

Table VI presents the outcomes of RNN LSTM model training, incorporating specific hyperparameters. The model employs 2 LSTM layers, each with 50 units and a dropout value of 0.2. Training involves the use of RNN with two optimizers, rmsprop and adam. This choice is informed by research [20] and [21], indicating that the Adam optimizer in LSTM models achieves the highest accuracy among various options, while research [22] highlights the superiority of the Rmsprop optimizer over alternatives like Adam and SGD. The model evaluation employs MSE, with the corresponding results detailed in Table VI.

TABLE VI. BEST HYPERPARAMETER AND MSE VALUE OF THE RNN LSTM MODEL FOR CPI

Country	Optimization	MSE
Brunei Darussalam	Adam	0.005
	RMSProp	0.006
Thailand	Adam	0.020
	RMSProp	0.019
Malaysia	Adam	0.018
	RMSProp	0.018
Indonesia	Adam	0.004
	RMSProp	0.004
Vietnam	Adam	0.005
	RMSProp	0.006

Tables VI and VII reveal experimental results for the RNN LSTM model with two optimizers, rmsprop and adam. The MSE values exhibit minimal differences between the two optimizers, whether applied to CPI or GDP datasets. Consequently, it can be inferred that the RNN-STM model with the Adam optimizer stands out as the superior choice for both inflation prediction using CPI and GDP, as indicated by the results in Tables VI and VII.

TABLE VII. BEST HYPERPARAMETER AND MSE VALUE OF THE RNN LSTM MODEL FOR GDP

Country	Optimization	MSE
Brunei Darussalam	Adam	0.012
	RMSProp	0.012
Thailand	Adam	0.046
	RMSProp	0.045
Malaysia	Adam	0.064
	RMSProp	0.067
Indonesia	Adam	0.022
	RMSProp	0.021
Vietnam	Adam	0.023
	RMSProp	0.027

IV. DISCUSSION

Upon evaluating the NN, SVM, and RNN LSTM models, it becomes evident that the NN model excels in predictive accuracy, boasting the smallest MSE values as depicted in Tables II and III. Notably, the NN model demonstrates greater stability in MSE results compared to the other two models. Examining Tables II and III, the optimal backpropagation NN model features 2 input layers with [64,32] neurons, employing ReLu activation functions. This model, particularly effective in predicting inflation using the Indonesian CPI dataset, achieves a remarkable MSE value of 0.001. Furthermore, an analysis of MSE results across Tables II, III, IV, V, VI, and VII reveals that, overall, these models consistently yield superior MSE values when predicting inflation using the CPI dataset, in contrast to the GDP dataset. This underscores the NN model's efficacy in inflation prediction, with its stable and superior performance across various scenarios.

TABLE VIII. PERFORMANCE COMPARISON WITH OTHER MODEL FROM PREVIOUS RESEARCH

Model	MSE
NN1	0.001
NN2	0.007
SVM1	0.001
SVM2	0.163
RNN LSTM1	0.004
RNN LSTM2	0.004

Table VIII contrasts the MAE values of the three models in this study with those from previous research. The NN model from our current study NN₁ outperforms its predecessor NN₂ [23], achieving an MSE value of 0.001 through improved hyperparameter selection and backpropagation methods. Notably, the chosen hyperparameters include [64, 32] neurons and [ReLU, ReLu] activation functions.

Similarly, the SVM model from our current study SVM₁ surpasses the one in the previous study SVM₂ [24], yielding an MSE value of 0.001. This improvement is attributed to refined hyperparameter selection, incorporating a sigmoid activation function, 10,000 C Value, and gamma 0.05.

Examining the RNN LSTM models, our current model RNN LSTM₁ outshines its predecessor RNN LSTM₂ [25], demonstrating its forecasting capabilities. The model excels with Adam optimization, a dropout value of 0.2 for both layers, and 50 LSTM units, showcasing its effectiveness in predicting outcomes.

V. CONCLUSION

The inflation prediction using three models NN Backpropagation, SVM, and RNN LSTM, utilizing two indicators, CPI and GDP values, yields promising results demonstrated by the MSE values. The best NN Backpropagation model achieves an outstanding MSE of 0.001 in predicting inflation using the CPI indicator in Indonesia, employing optimal hyperparameters of [64, 32] for the number of neurons and [ReLU, ReLU] for the activation function. Meanwhile, the best SVM model achieves an MSE of 0.001 in predicting inflation using CPI indicators in Malaysia and Thailand, with distinct hyperparameters for each country. The best RNN LSTM model produces an MSE of 0.004 for predicting inflation values using CPI indicators in Indonesia, employing the best hyperparameters, 0.2 dropout value, 50 LSTM units, 2 hidden layers, and the Adam optimization function.

The study highlights that these three models outperform their predecessors, especially in predicting inflation values. Moreover, the results underscore the significance of hyperparameter selection and dataset choice, influencing accuracy variations. Notably, Fig. 5, 6, 7, 8, 9 illustrate the NN Backpropagation model's performance on training data for forecasting inflation using CPI values. The data reveals a decrease in inflation during the 2020 pandemic, followed by a resurgence in 2022 across Indonesia, Malaysia, Brunei Darussalam, Thailand, and Vietnam. The observed correlation between CPI values and inflation during the COVID-19 period offers valuable insights for post-pandemic strategies to stabilize inflation rates in these countries. Future research could explore more intricate models or alternative parameter-tuning techniques to enhance the predictive performance of these models for CPI and GDP indicators.

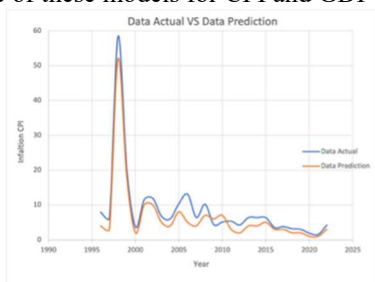


Fig. 5. Result training data NN backpropagation CPI Indonesia.

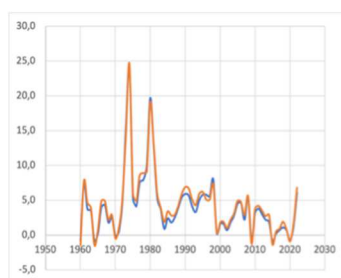


Fig. 6. Result training data NN backpropagation CPI Thailand.

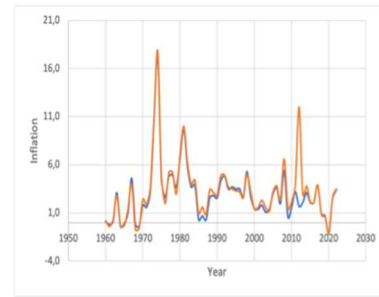


Fig. 7. Result training data NN backpropagation CPI Malaysia.

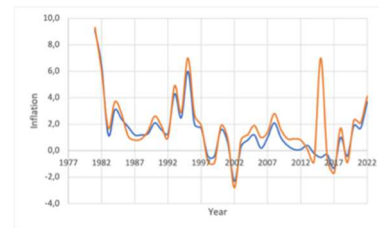


Fig. 8. Result training data NN backpropagation CPI Brunei Darussalam.

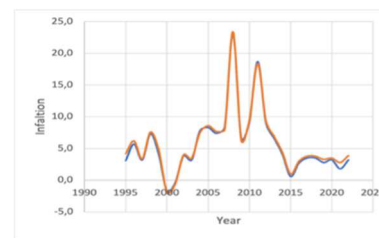


Fig. 9. Result training data NN backpropagation CPI Vietnam.

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